

A3 L5 absolute scale framework v0.1 — what sets the SM absolute mass scale in BST? Four candidate sources surveyed (Planck $\times \alpha^n$, Λ -tied, Higgs VEV, Koons tick), each rejected with explicit numerical evidence: $m_e/m_{\text{Planck}} \approx \alpha^{10.47}$, NOT a clean substrate integer. HONEST CONCLUSION: BST derives mass RATIOS rigorously but lacks an absolute-scale derivation in current scope; L5 is genuinely OPEN and multi-month research.

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A3 L5 absolute scale framework v0.1

0. The question

BST derives mass RATIOS extensively: - $m_\mu/m_e = (24/\pi^2)^6 \approx 206.77$ (T190, 0.004% precision) - $m_\tau/m_e = 49.71$ (T2003, ~0.05%) - $m_p/m_e = 6\pi^5 = C_2 \cdot \pi^{(n_C)}$ (T187, 0.002%) - $m_W/m_Z = \sqrt{(g/N_c^2)}$ (Weinberg derived) - $\sin^2\theta_W = \text{rank}/N_c^2 = 2/9$ - PMNS angles = substrate-primary ratios over N_{max}

But the ABSOLUTE scale (e.g., $m_e \approx 0.511$ MeV; Higgs VEV ≈ 246 GeV; cosmological constant Λ scale) is not yet substrate-derived. What candidates exist? Why doesn't BST yet have an absolute scale?

1. Four candidate sources (each rejected)

Candidate (a) — $m_e = m_{\text{Planck}} \times \alpha^n$, n = substrate-natural integer

$$m_e / m_{\text{Planck}} = 4.185 \times 10^{-23} \alpha^n = 4.185 \times 10^{-23} \rightarrow n \approx 10.473$$

This is NOT a clean substrate-natural integer (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$, $N_{\text{max}}=137$ — none of these give 10.47 directly, and integer combinations don't either). Closest: $n = 10$ (= dim adjoint of $\text{so}(5) = C_2 \cdot \text{rank} = \dim \mathfrak{p}$ Cartan) gives $m_e \approx 5.23$ MeV, off by 10 $\times$ from observed.

Rejected: no clean substrate-integer exponent works.

Candidate (b) — Cosmological constant Λ -tied

$$\Lambda^{(1/4)} \approx 3.86 \times 10^{-12} \text{ GeV} \approx 3.86 \text{ meV} (\Lambda \text{ in Planck units} \approx 10^{-122}) \quad m_e / \Lambda^{(1/4)} \approx 1.32 \times 10^8$$

Massive ratio — m_e is $\sim 10^8$ times the cosmological scale. No direct scale match.

Rejected: Λ -tied scale is too small to anchor m_e .

Candidate (c) — Higgs VEV substrate-derivation

$v \approx 246$ GeV (electroweak scale, sets all SM fermion masses via Yukawa couplings) $m_e = y_e \cdot v$ with $y_e \approx 2.1 \times 10^{-6}$

For BST to derive m_e absolutely, it would need to derive BOTH v AND y_e from substrate primaries. Neither is currently derivable from the BST integers.

Rejected: not in current BST scope; would require an entirely separate substrate-EWSB derivation.

Candidate (d) — Koons tick $t_K = t_{\text{Planck}} \times \alpha^{(C_2^2)}$

$t_K \approx 6.39 \times 10^{-121}$ s (substrate's sub-Planck tick scale) $1/t_K \approx 1.57 \times 10^{120}$ Hz (energy scale)

Vastly above SM scales (which are at most $\sim 10^{25}$ Hz for the electroweak scale). Koons tick is the substrate's "fastest" scale, not anchoring SM masses.

Rejected: wrong order of magnitude.

2. The structural asymmetry

BST derives mass RATIOS via substrate-primary closed forms (T190, T2003, T187, Weinberg, PMNS — all clean). But the SCALE is an external input — typically the electron mass $m_e \approx 0.511$ MeV is taken as observed, and all other masses are predicted as ratios.

Why this asymmetry? Several structural reasons:

- BST primaries are dimensionless integers (rank, N_c , n_C , C_2 , g , N_{max}). Without a dimensionful input, no absolute scale can emerge.
- D_{IV}^5 is a mathematical bounded symmetric domain — has no inherent physical "size."
- Bergman kernel is dimensionless — $c_{FK} = 225/\pi^{(9/2)}$ is a normalization factor, dimensionless.
- Mass mechanism is RATIO-multiplicative: lepton at Casimir $5/2 = \rho_1$ = kernel exponent gives the natural unit, but the unit itself is not predicted from the substrate without external anchor.

So the absolute scale requires either: (i) An EXTERNAL dimensionful input (electron mass, Planck scale, etc.). (ii) A NEW substrate-natural mechanism that ties a substrate primary to a physical scale — possibly via the cosmological constant Λ (BST-derived) or via some renormalization-group flow from the Planck scale.

3. The genuinely-open candidates for v0.2+ research

If we want to DERIVE the absolute scale, the most promising directions:

(e) $\Lambda \rightarrow$ SM scale via substrate cosmological evolution

If Λ is BST-derived (T180 cosmological constant Λ Zone-4 substrate-mechanism, Tuesday derivation), and Λ sets the cosmological scale, then the SM scale might emerge via running from Λ through substrate evolution. The hierarchy $m_e / \Lambda^{(1/4)} \sim 10^8$ could be a structural ratio.

Multi-week: trace the substrate's cosmological evolution and check if SM scale falls out at a specific epoch.

(f) Higgs VEV from Bergman normalization constant

The Bergman kernel has $c_{FK} = 225/\pi^{(9/2)}$ (dimensionless). If the substrate's natural "field strength" at the kernel singularity sets a scale via Hilbert-space normalization, the Higgs VEV might be derivable.

Multi-week: explicit Bergman / Hardy integral computation for the substrate ground state, identify any dimensional anchor.

(g) Mersenne-tower / $N_{\max}$ -tied scale

$N_{\max} = 137$ anchors $\alpha (= 1/N_{\max})$. A natural scale via $N_{\max}$ would be $(M_{\{N_{\max}\}})^{(1/k)}$ for some k , but M_{137} is astronomically large. Alternative: a specific Mersenne residue (e.g., the largest $M_p < N_{\max}$ with p prime) might give a substrate-natural scale.

Multi-week: check if any Mersenne residue gives the EWSB or m_e scale.

4. Conclusion (honest)

L5 absolute scale is genuinely OPEN in BST. The framework currently: - Derives RATIOS cleanly (the strongest of BST's successes). - Does NOT derive an absolute scale (the weakest of BST's gaps). - The structural asymmetry — dimensionless substrate primaries cannot produce a dimensionful scale without an external anchor.

v0.1 framework deliverable: - Identify the question clearly. - Survey 4 candidate sources (Planck $\times \alpha^n$, Λ -tied, Higgs VEV, Koons tick) — all rejected with numerical evidence. - Identify 3 directions for v0.2+ research (Λ -cosmological-evolution; Bergman-normalization-Higgs; Mersenne-residue). - Honest disposition: this is multi-month research.

5. Honest scope + tier

RIGOROUS (existing BST results): mass RATIOS (T187, T190, T2003, Weinberg, PMNS); BST primaries are dimensionless integers; Bergman kernel is dimensionless.

FRAMEWORK (this v0.1): maps the absolute-scale question; surveys candidate sources; identifies open directions.

OPEN (multi-month research): actual derivation of an absolute scale from substrate primaries — requires either a new dimensionful mechanism (cosmological-evolution, Bergman-normalization, Mersenne-residue) or recognition that BST is intrinsically RATIO-only and the absolute scale is an external observation.

Cal #27 / honesty: v0.1 is a NEGATIVE FRAMEWORK — it surveys candidate sources and finds none directly works, then identifies the structural asymmetry (BST ratios vs no absolute scale). This is honest about a real limitation. NOT a derivation; explicitly flagged as multi-month open work.

Routed: → Keeper: L5 absolute scale is genuinely OPEN; honest framework filed. → Casey: this is a real BST limitation — substrate is dimensionless integers; absolute scale requires external anchor (electron mass observed) OR a new substrate-cosmological-evolution mechanism (multi-month research). → me: continuing to Lyra Queue #4 = B4 MTC from R-matrix.

— Lyra, A3 L5 absolute scale framework v0.1. The question: what sets the SM absolute mass scale in BST? Four candidate sources surveyed: (a) $m_{\text{Planck}} \times \alpha^n$ with n =substrate-integer — REJECTED ($n=10.47$ not integer); (b) Λ -tied — REJECTED ($m_e/\Lambda^{1/4} \approx 10^8$ no match); (c) Higgs VEV derivation — out of scope (would need v AND y_e from substrate); (d) Koons tick — wrong order of magnitude (10^{120} Hz). HONEST CONCLUSION: BST derives ratios cleanly but lacks absolute scale; v0.2+ directions are Λ -cosmological-evolution, Bergman-normalization-Higgs, Mersenne-residue. Multi-month research; framework, not derivation.